

AI Supply Chain

A beginner and Taiwan-specific exploration

Why this topic first?

- Understand the basic terms that commonly use in AI
- Understand where does Taiwan's semiconductor locate in the AI supply chain
- Serve as a beginning of thinking what role to play in AI
- Future investment consideration

Model

Description: Transform data with algorithms to compute (tokens)

Components:

- Training
- Inference

Main Players: Frontier AI lab

- USA: OpenAI, Anthropic, Grok, Google (?)
- European: Mistral
- China: Deepseek, Alibaba (Qwen)

Data center

Descriptions:

- The infrastructure to develop and maintain AI models

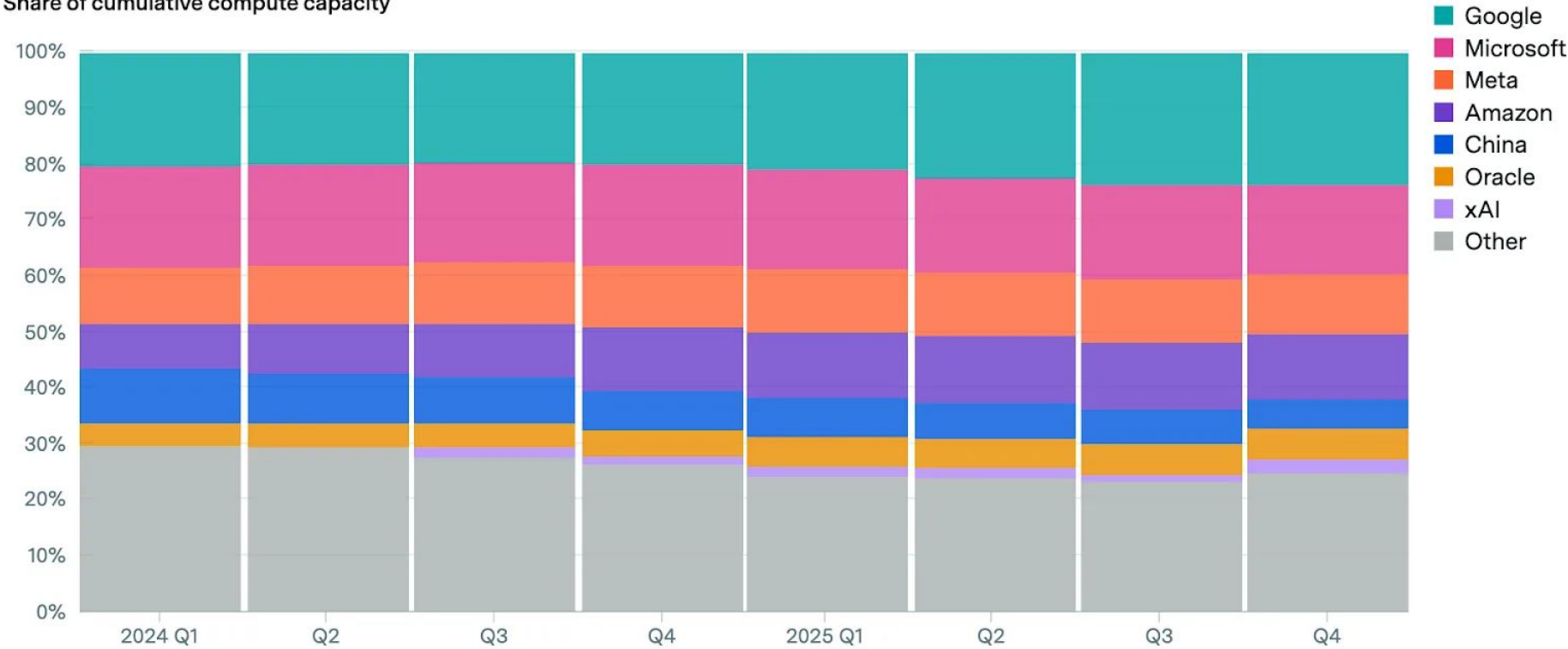
Components:

- Advanced chips (GPUs, TPUs, etc.)
- Server racks
- Electricity

Main players: Google, Amazon, Microsoft, Meta

AI chip ownership over time

Share of cumulative compute capacity



Nvidia cumulative totals include sales from 2022 onward. TPU cumulative totals include sales from 2023 onward.

Hardware manufacturing (Where Taiwan shine !!)

GPU production:

- **design** → **foundry (fabrication)** → **advanced packaging**

Design: Nvidia, AMD, Intel

Foundry:

- TSMC (Compute/Logic)
- SK Hynix, Samsung, Micron (HBM) High-bandwidth memory

Advanced packaging: TSMC, ASE

Hardware manufacturing (Where Taiwan shine !!)

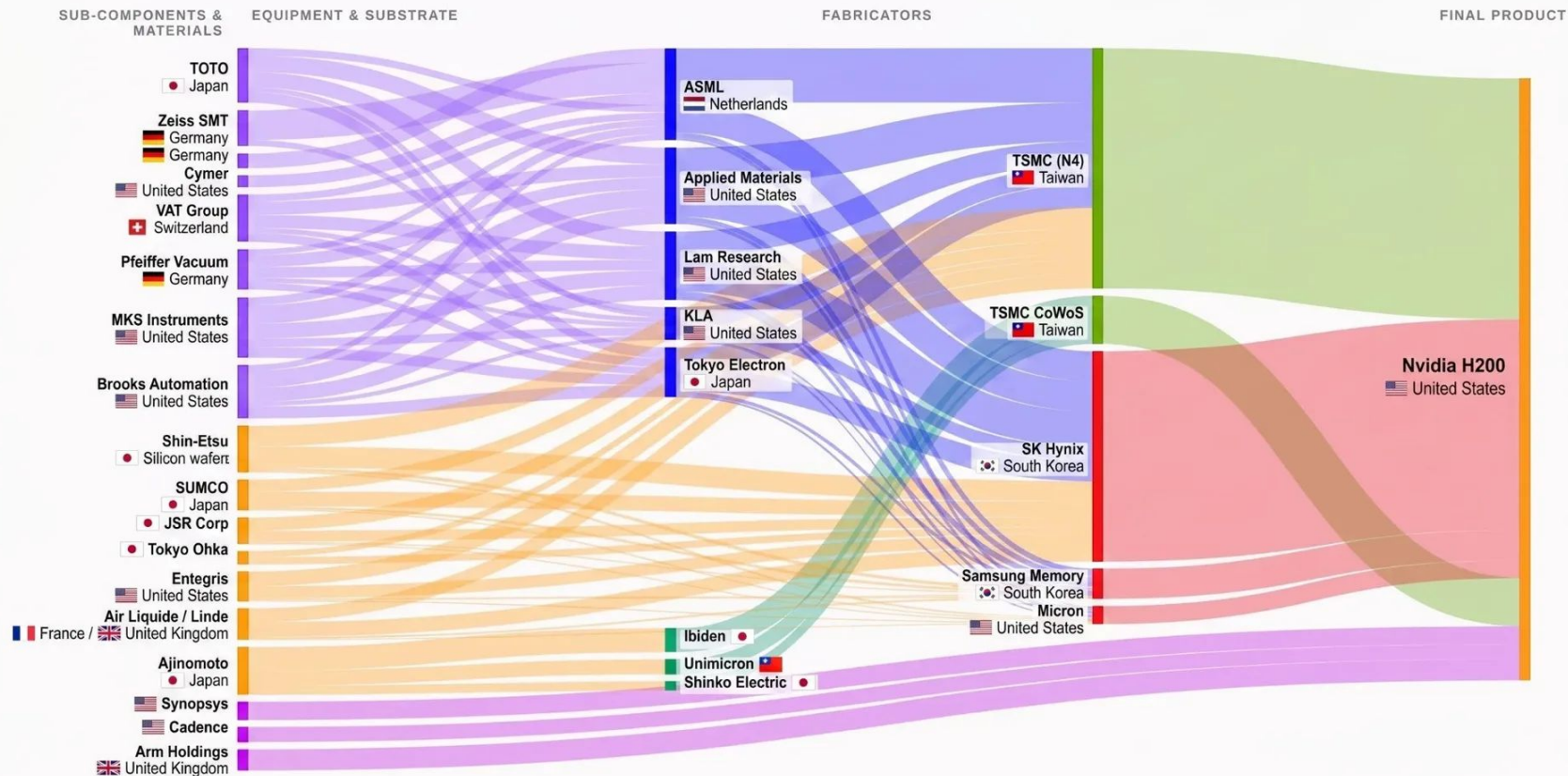
AI Server assembly

- Accelerators: GPU, TPU, NPU
- CPU
- RAM
- SSD
- Power and cooling
- high-speed networking

Main players: Foxconn, Wistron, Quanta, Compal, Inventec

The supply chain inside one Nvidia H200

31 suppliers across 4 tiers, weighted by proportional spend per chip. Sub-components and materials feed 5 equipment makers and 3 substrate assemblers, which serve 5 fabricators producing the GPU die, CoWoS packaging, and HBM3e memory stacks.



Homeworks

- Get familiar with technical terms
- What roles should I play in the game?
- Start to think about topics that demonstrate your understanding of the industry and things to talk in interview
- Find some readings want to share
- Next Session: Salary

References

- <https://weightsandwords.com/a-beginners-guide-to-the-ai-supply-chain>
- <https://www.makesureiknowit.com/blog/who-supplies-nvidia>

Recent discussion:

- <https://www.youtube.com/watch?v=VA4v0bl7npQ>